Project plan

SignFlow — Digital Document Approval Workflow for a University

1. Planning basis

The case gives per-file-type effort of 3 days for workflow design, 4 days for configuration, and 2 days for testing. The schedule below uses **elapsed working days as a planning assumption** so dependencies can be shown. It is not a promise of delivery time. The plan assumes a small team can work on independent activities in parallel and that the university supplies rules and reviewers when needed.

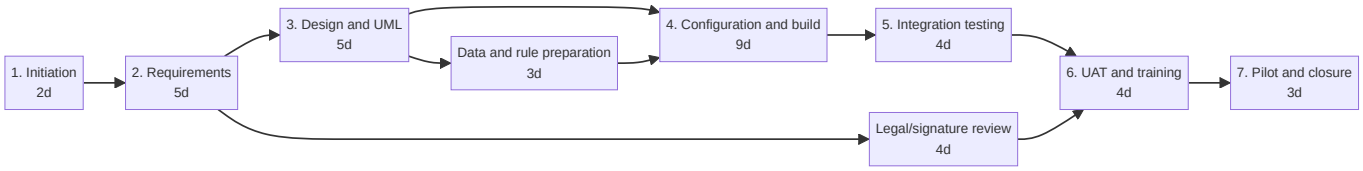
The recommended first release covers purchase requests, leave, and event approvals. These are 1,600 of the 2,000 monthly files, or 80% of volume.

2. Work breakdown structure

WBS	Work package	Main output
1.0	Initiation and governance	Charter, stakeholders, decision log
1.1	Confirm scope and release boundary	Top-three scope decision
1.2	Confirm approval owners	Named Registrar, IT, legal, and business owners
2.0	Requirements	Validated SRS
2.1	Questionnaires and interviews	Elicitation notes
2.2	Prototype walkthroughs	Usability and policy feedback
2.3	FR/NFR and MoSCoW baseline	Requirements baseline
3.0	Analysis and design	UML package and configured rules
3.1	Workflow and routing rules	File-type and amount decision rules
3.2	UML diagrams	Consistent UML package
3.3	Signature and audit design	Integration and audit design
4.0	Build/configure	Configured SignFlow release
4.1	Intake and file record	File creation and validation
4.2	Routing and inbox	Route and approval steps
4.3	Signature, reminders, audit	Supporting services
5.0	Verification	Test evidence and defect log
5.1	Functional and integration testing	Test results
5.2	Security, audit, and signature checks	NFR evidence
5.3	User acceptance testing	Signed UAT result
6.0	Pilot and closure	Pilot report and closure note
6.1	Training and assisted onboarding	Training record
6.2	Pilot launch	Pilot metrics
6.3	Review, fixes, and release decision	Closure or change plan

3. Network logic and critical path

The following is a dependency network. Durations are planning assumptions in working days.



Critical path assumption: A → B → C → D → E → F → G = 2 + 5 + 5 + 9 + 4 + 4 + 3 = **32 working days**. The legal/signature review and data/rule preparation are shown as parallel branches. If either branch takes longer than its successor activity, the critical path must be recalculated.

4. Gantt-style schedule

Activity	Start day	Finish day	Dependency	Milestone/output
Initiation	1	2	—	M1: scope decision
Requirements	3	7	Initiation	M2: SRS baseline
Design and UML	8	12	Requirements	M3: design review
Legal/signature review	8	11	Requirements	Signature method decision
Data and rule preparation	13	15	Design	Rules ready
Configuration and build	13	21	Design; rules can overlap after day 12	M4: configured build
Integration testing	22	25	Build	Test evidence and defects
UAT and training	26	29	Integration testing; legal review	M5: UAT decision
Pilot and closure	30	32	UAT	M6: pilot release/closure

5. Resource allocation assumption

The case does not supply staffing. This indicative allocation shows responsibilities rather than headcount commitments.

Role	Main allocation	Key approvals
Registrar	Sponsor and decision owner	Scope, acceptance, escalations
IT head / project lead	Planning, architecture, release	Technical design and release
Business analyst	Elicitation, SRS, traceability	Requirements baseline
Workflow/configuration developer	Rules, screens, integrations	Build readiness
QA engineer	Test design, execution, metrics	Test exit recommendation
Administration representatives	Process knowledge and UAT	Business acceptance
Senior officers	Walkthroughs and UAT	Usability feedback
Legal/compliance representative	Signature validity review	Legal approval

6. Monitoring and control

Phase	Metric	Target	Frequency	Escalation threshold	Corrective action	Approver
Requirements	Baseline completion	100% Must requirements reviewed	Twice weekly	Any Must requirement without owner at phase exit	Schedule owner workshop; update baseline	Registrar and IT head
Design	Rule coverage	All top-three file types have approved routing rules	Weekly	Any type or amount band without a rule	Add rule review and test cases	IT head and business owner
Build	Workflow readiness	All Must flows demonstrable	Weekly	Any Must flow blocked at build exit	Reprioritise defects or defer only with approval	IT head
Testing	Test pass rate	100% Must tests pass	Daily during test	Any open critical defect or failed legal/audit test	Stop release; fix and rerun	IT head and QA lead
Pilot	Average approval time	Under 4 days	After each reporting period	Average at or above 4 days	Analyse queue and routing; adjust staffing/rules	Registrar
Pilot	Lost-file exceptions	Zero	Daily	One unretrievable submitted file	Freeze affected route, restore/reconcile, investigate	IT head and Registrar

For actual-vs-plan control, record `variance = actual finish day - planned finish day`. A positive value means delay. The phase owner proposes corrective action; the listed approver decides whether to accept a revised date or change scope.